

A.Y.-2023-24 – SY BTech - Department of Computer Sciences and Engineering	
ODD SEMESTER (SEM-III)	
Subject: Mathematics for Intelligent System	Code: 22BSCS3010T
Term Test – I	
Unit -2 and Unit - 3	

Que. No.	Question	Marks	CO	Level
Unit-2				
1	Let V be the set of ordered pairs (x, y) of complex numbers. Show that V is a vector space over the real field R with addition in V and scalar multiplication on V defined by $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$ $k(x_1, y_1) = (kx_1, ky_1)$	10	CO2	L2
2	Determine whether the following function is linear transformation i) $T: P_2 \rightarrow P_2, \quad T(a_0 + a_1x + a_2x^2) = a_0x + a_1x^2 + a_2x^3$ ii) $T: M_{22} \rightarrow R, \quad T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a^2 + b^2$	10	CO2	L2
3	Determine whether the following set of vectors form a basis for V i) (1,1,1), (1,2,3), (2, -1, 1) $V = R^3$ ii) $-4 + x + 3x^2, 6 + 5x + 2x^2, 8 + 4x + x^2$	10	CO2	L2
4	Show that the set V of all pairs of real numbers of the form (1, x) with the operation defined as $(1, x) + (1, y) = (1, x + y)$ and $k(1, x) = (1, kx)$ Is a vector space	10	CO2	L2
5	Let $V = R^3$, show that W is a subspace of V where a) $W = \{(a, b, 0): a, b \in R\}$ b) $W = \{(a, b, c): a + b + c = 0\}$	05	CO2	L2
6	Determine the linear dependence and independence of the vectors (1,1,2,1), (1,0,0,2), (4,6,8,6), (0,3,2,1)	05	CO2	L2
7	Determine whether the following matrices span M_{22} $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$	05	CO2	L2
8	Find the basis for the orthogonal complement of the subspace W of R^3 span by vectors (2, 0, -1), (4, 0, -2)	05	CO2	L2
Unit-3				
9	Find the eigenvalues and Eigen vectors of the following matrices. Are these matrices diagonalisable. If so find the modal (transforming) matrix and the diagonal matrix. $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$	10	CO3	L3
10	If $A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \\ 1 & 3 \\ 2 & 2 \end{bmatrix}$, Find e^A and 4^A	10	CO3	L3
11	Verify Caley Hamilton Theorem and also find $A^9 - 6A^8 + 10A^7 - 3A^6 + A + I$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$	10	CO3	L3

12	Find the eigenvalues and Eigen vectors of the following matrix. Is the matrix diagonalisable? If so find the modal (transforming) matrix and the diagonal matrix. $A = \begin{bmatrix} -9 & 2 & 6 \\ 5 & 0 & -3 \\ -16 & 4 & 11 \end{bmatrix}$	10	CO3	L3
13	If $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, find the eigenvalues of the following matrices: $A^T, A^{-1}, A^2 - 2A + I, adj(A)$	05	CO3	L3
14	Check whether the matrix A derogatory? If yes, find its minimal polynomial. $A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & 4 \end{bmatrix}$	05	CO3	L3
15	Find the eigenvalues and eigenvectors of $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$	05	CO3	L3
16	Verify Cayley-Hamilton Theorem and hence find A^{-1} for where $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$	05	CO3	L3